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November 2025

## 1 Data Preparation

The dataset integrates multiple financial time series into a unified monthly panel for the purpose of modeling the **Neural Beta** across three dimensions of systematic exposure: energy equities (**VDE**), fiat currency strength (**FIAT**), and the broad equity market (**Mkt--RF**). This consolidated data structure enables the model to estimate dynamic betas that jointly reflect sensitivity to sectoral, macroeconomic, and market-level risk drivers.

Each observation is indexed by year and month, and contains log returns for fiat currency, gold, energy equities (**VDE**), the market portfolio, and cryptocurrency assets. Standard Fama–French factors (**Mkt--RF**, **SMB**, **HML**, **RF**) are incorporated to capture classical linear risk premia, which serve as theoretical anchors for the nonlinear neural framework. In addition to raw returns, rolling-window features—including means, standard deviations, and z-scores—are computed for each variable, as well as a rolling correlation with the market (**corr\_market\_w**). These transformations produce a temporally coherent and feature-rich dataset suitable for learning time-varying relationships between asset returns and their underlying factor sensitivities.

### Technical Computations

For each return series  $r_t$ , a rolling window of size  $w$  months is applied to compute the moving average  $\mu_t$  and standard deviation  $\sigma_t$ :

$$\mu_t = \frac{1}{w} \sum_{i=t-w+1}^t r_i, \quad \sigma_t = \sqrt{\frac{1}{w-1} \sum_{i=t-w+1}^t (r_i - \mu_t)^2}.$$

The standardized (z-score) representation is defined as:

$$z_t = \frac{r_t - \mu_t}{\sigma_t},$$

providing a normalized measure of each asset’s deviation from its recent trend. To capture dynamic comovements, a rolling correlation is also computed between each asset and the equity market return (**vwret**):

$$\rho_t = \text{corr}(r_{t-w+1:t}, \text{vwret}_{t-w+1:t}).$$

These engineered features collectively encode short-term volatility, momentum, and market linkage effects that serve as key explanatory signals for predicting the Neural Betas against VDE, FIAT, and Equity.

## 2 Neural Beta Model

The **Neural Beta** framework employs a simple feedforward neural network to estimate dynamic factor loadings across three major systematic exposures: energy equities (**VDE**), fiat currency (**FIAT**), and the broad equity market (**Equity**). Unlike traditional linear regression models, which assume constant betas, this neural approach allows  $\beta_t$  to vary over time as a nonlinear function of observed features, capturing changing market sensitivities and interactions between factors.

### Architecture

The model is implemented as a lightweight multilayer perceptron (MLP) with one hidden layer and a single output node representing the predicted beta. A nonlinear activation function, such as ReLU or Tanh, is applied between layers to enable the network to learn complex relationships in the data. This compact design maintains interpretability while allowing the model to adapt to evolving market conditions.

### Loss Function

Training is guided by a root-mean-squared error (RMSE) objective that links predicted betas to realized returns. For each time period, the model predicts a beta  $\hat{\beta}$ , which is multiplied by the observed factor return  $r_{t+1}^{(asset)}$  to estimate the asset’s return. The loss function is defined as:

$$\mathcal{L}(\theta) = \sqrt{E \left[ (r_{t+1} - \hat{\beta}_{t+1} r_{t+1}^{(asset)})^2 \right] + \epsilon},$$

where  $\epsilon$  is a small constant added for numerical stability. This setup embeds the intuition of the CAPM within a differentiable framework, allowing the network to learn smooth, time-varying betas directly from data.

## 3 Hyperparameter Configuration and Results

Model performance was evaluated across several hyperparameter settings to identify the combination that best balanced training stability and validation accuracy. The configuration yielding the lowest validation loss used a window size of 3, a hidden dimension of 8, a learning rate of 0.001, and a sigmoid activation function:

*Bestconfiguration* : {window = 3, hidden\_dim = 8, learning\_rate = 0.001, activation = sigmoid}.

This setup achieved a validation RMSE of approximately 0.227, indicating a strong fit relative to other tested configurations.

## Training Dynamics

Figure 1 illustrates the evolution of the training and validation RMSE over 50 epochs. The training loss exhibits small fluctuations but remains stable, while the validation loss stays consistently lower and flat, suggesting good generalization and an absence of overfitting.

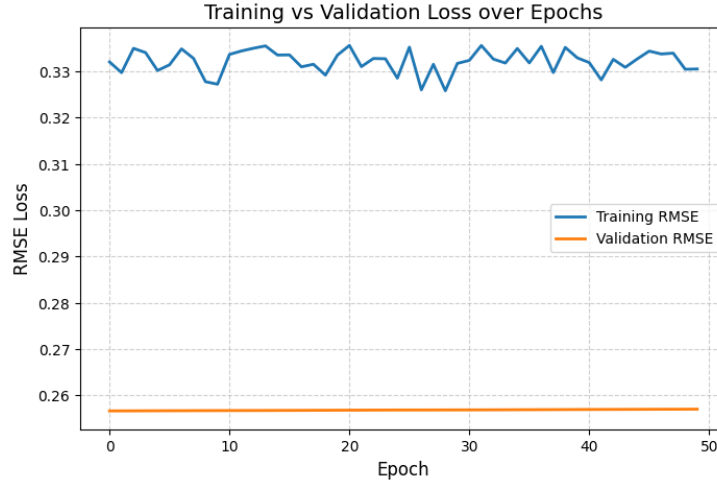


Figure 1: Training vs. Validation RMSE over Epochs

## 4 Results

This section summarizes the distribution of the predicted Neural Betas for each factor exposure. The first table reports results from the baseline model trained without Fama–French factors, while the second table includes those factors to evaluate their effect on estimated betas. Each table presents descriptive statistics across the testing period, including count, mean, standard deviation, and percentile ranges.

## 5 Results

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## Neural Betas Without Fama–French Factors

Table 1: Summary statistics of Neural Betas without Fama–French factors.

|             | Count | Mean    | Std    | Min     | 25%     | 50%     | 75%     | Max     |
|-------------|-------|---------|--------|---------|---------|---------|---------|---------|
| Energy Beta | 128   | -0.1580 | 0.0120 | -0.2019 | -0.1655 | -0.1582 | -0.1497 | -0.1292 |
| Fiat Beta   | 128   | -0.8136 | 0.0112 | -0.8440 | -0.8218 | -0.8134 | -0.8063 | -0.7843 |
| Equity Beta | 128   | 0.0533  | 0.0100 | 0.0283  | 0.0465  | 0.0536  | 0.0598  | 0.0903  |

## Neural Betas With Fama–French Factors

Table 2: Summary statistics of Neural Betas with Fama–French factors.

|                    | Count | Mean    | Std    | Min     | 25%     | 50%     | 75%     | Max    |
|--------------------|-------|---------|--------|---------|---------|---------|---------|--------|
| Energy Beta (Fama) | 128   | -0.1494 | 0.0747 | -0.2806 | -0.2021 | -0.1585 | -0.1009 | 0.0049 |
| Fiat Beta (Fama)   | 128   | 0.0938  | 0.3322 | -0.4421 | -0.1260 | 0.1074  | 0.3359  | 0.6945 |
| Equity Beta (Fama) | 128   | 0.2360  | 0.0906 | 0.0436  | 0.1647  | 0.2338  | 0.3077  | 0.4014 |

## Effect of Adding Fama–French Factors on Betas

- **Energy beta:** Became slightly less negative ( $-0.158 \rightarrow -0.149$ ) and far more variable (standard deviation increased from 0.01 to 0.07), indicating that the Fama factors absorbed part of the common energy-related risk.
- **Fiat beta:** Flipped from strongly negative ( $-0.81$ ) to mildly positive ( $+0.09$ ) with high dispersion, suggesting that its earlier signal overlapped heavily with broader market factors.
- **Equity beta:** Increased substantially ( $0.05 \rightarrow 0.24$ ) and became more volatile, revealing a stronger underlying connection to equity markets once systematic effects were controlled.
- **Overall:** Adding Fama–French factors increased dispersion and clarified distinct sources of risk across assets.

## 6 Analysis

To assess how Neural Betas relate to realized performance, assets were grouped into quartiles under two perspectives: (1) by their estimated betas, and (2) by their realized returns. For each quartile, the mean beta and mean excess return (or mean return) were computed, along with the spread between the highest and lowest quartiles (Q4–Q1). This dual analysis provides insight into whether higher-beta cryptocurrencies tend to earn higher returns, and how these relationships vary across energy, fiat, and equity exposures.

### Asset Class: Energy (Quartiles by Beta, Mean Beta, Mean Excess Return)

Table 3: Quartile analysis of Neural Betas and excess returns — Energy (by Beta).

| Quartile | Crypto                         | Mean Beta | Mean Excess Return |
|----------|--------------------------------|-----------|--------------------|
| Q1       | LC_log_return                  | -0.158    | -0.0400            |
| Q2       | ETH_log_return                 | -0.158    | 0.0063             |
| Q3       | BPC_log_return                 | -0.157    | -0.0328            |
| Q4       | BTC_log_return                 | -0.158    | -0.0146            |
| Q4 – Q1  | BTC_log_return – LC_log_return | -0.000    | 0.0254             |

### Asset Class: Energy (Quartiles by Return, Mean Beta, Mean Return)

Table 4: Quartile analysis of Neural Betas and returns — Energy (by Return).

| Quartile (by Mean Return) | Crypto                         | Mean Beta | Mean Return |
|---------------------------|--------------------------------|-----------|-------------|
| Q1                        | LC_log_return                  | -0.158    | -0.0149     |
| Q2                        | BPC_log_return                 | -0.157    | -0.0077     |
| Q3                        | BTC_log_return                 | -0.158    | 0.0105      |
| Q4                        | ETH_log_return                 | -0.158    | 0.0313      |
| Q4 – Q1                   | ETH_log_return – LC_log_return | -0.000    | 0.0462      |

Across energy-linked exposures, Neural Betas remain nearly constant, indicating uniform sensitivity to the energy factor. The Q4–Q1 spread shows a modest increase in return with no meaningful change in beta, suggesting that performance differences arise from idiosyncratic factors rather than systematic beta exposure.

**Asset Class: Fiat (Quartiles by Beta, Mean Beta, Mean Excess Return)**

Table 5: Quartile analysis of Neural Betas and excess returns — Fiat (by Beta).

| Quartile | Crypto                         | Mean Beta | Mean Excess Return |
|----------|--------------------------------|-----------|--------------------|
| Q1       | ETH.log_return                 | -0.816    | 0.0295             |
| Q2       | BTC.log_return                 | -0.814    | 0.0087             |
| Q3       | BPC.log_return                 | -0.810    | -0.0095            |
| Q4       | LC.log_return                  | -0.813    | -0.0167            |
| Q4 – Q1  | LC.log_return – ETH.log_return | 0.004     | -0.0462            |

**Asset Class: Fiat (Quartiles by Return, Mean Beta, Mean Return)**

Table 6: Quartile analysis of Neural Betas and returns — Fiat (by Return).

| Quartile (by Mean Return) | Crypto                         | Mean Beta | Mean Return |
|---------------------------|--------------------------------|-----------|-------------|
| Q1                        | LC.log_return                  | -0.813    | -0.0149     |
| Q2                        | BPC.log_return                 | -0.810    | -0.0077     |
| Q3                        | BTC.log_return                 | -0.814    | 0.0105      |
| Q4                        | ETH.log_return                 | -0.816    | 0.0313      |
| Q4 – Q1                   | ETH.log_return – LC.log_return | -0.004    | 0.0462      |

Under fiat exposure, all Neural Betas are negative, confirming the inverse link between crypto assets and fiat strength. However, returns increase steadily from Q1 to Q4 despite negligible beta variation. This pattern suggests that high-return cryptos are not necessarily driven by stronger fiat sensitivities, but rather by independent or speculative factors.

**Asset Class: Equity (Quartiles by Beta, Mean Beta, Mean Excess Return)**

**Asset Class: Equity (Quartiles by Return, Mean Beta, Mean Return)**

For equity exposure, both beta and return increase slightly from Q1 to Q4. Although the differences are small, they align with the classical risk–return intuition that assets with higher equity sensitivity earn higher excess returns during favorable market conditions.

Table 7: Quartile analysis of Neural Betas and excess returns — Equity (by Beta).

| Quartile | Crypto                         | Mean Beta | Mean Excess Return |
|----------|--------------------------------|-----------|--------------------|
| Q1       | LC.log_return                  | 0.053     | -0.0220            |
| Q2       | BPC.log_return                 | 0.054     | -0.0148            |
| Q3       | BTC.log_return                 | 0.053     | 0.0034             |
| Q4       | ETH.log_return                 | 0.054     | 0.0242             |
| Q4 – Q1  | ETH.log_return – LC.log_return | 0.000     | 0.0462             |

Table 8: Quartile analysis of Neural Betas and returns — Equity (by Return).

| Quartile (by Mean Return) | Crypto                         | Mean Beta | Mean Return |
|---------------------------|--------------------------------|-----------|-------------|
| Q1                        | LC.log_return                  | 0.053     | -0.0149     |
| Q2                        | BPC.log_return                 | 0.054     | -0.0077     |
| Q3                        | BTC.log_return                 | 0.053     | 0.0105      |
| Q4                        | ETH.log_return                 | 0.054     | 0.0313      |
| Q4 – Q1                   | ETH.log_return – LC.log_return | 0.000     | 0.0462      |

## Do High- $\beta$ Cryptos Deliver Higher Returns?

Across all three asset classes, the relationship between Neural Beta and realized return appears weak. Betas are generally stable and show little separation across quartiles, indicating that most variation in returns is driven by idiosyncratic rather than systematic factors. In the equity dimension, a modest positive association emerges—higher- $\beta$  cryptos earn slightly higher returns—but in energy and fiat dimensions, no clear premium is observed. Overall, the results suggest that within this sample, high- $\beta$  cryptocurrencies do not consistently deliver higher excess returns than low- $\beta$  ones, implying that neural betas capture exposure rather than reward.